MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, STRAIGHT-THRU, HERMAPHRODITIC CONTACT FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

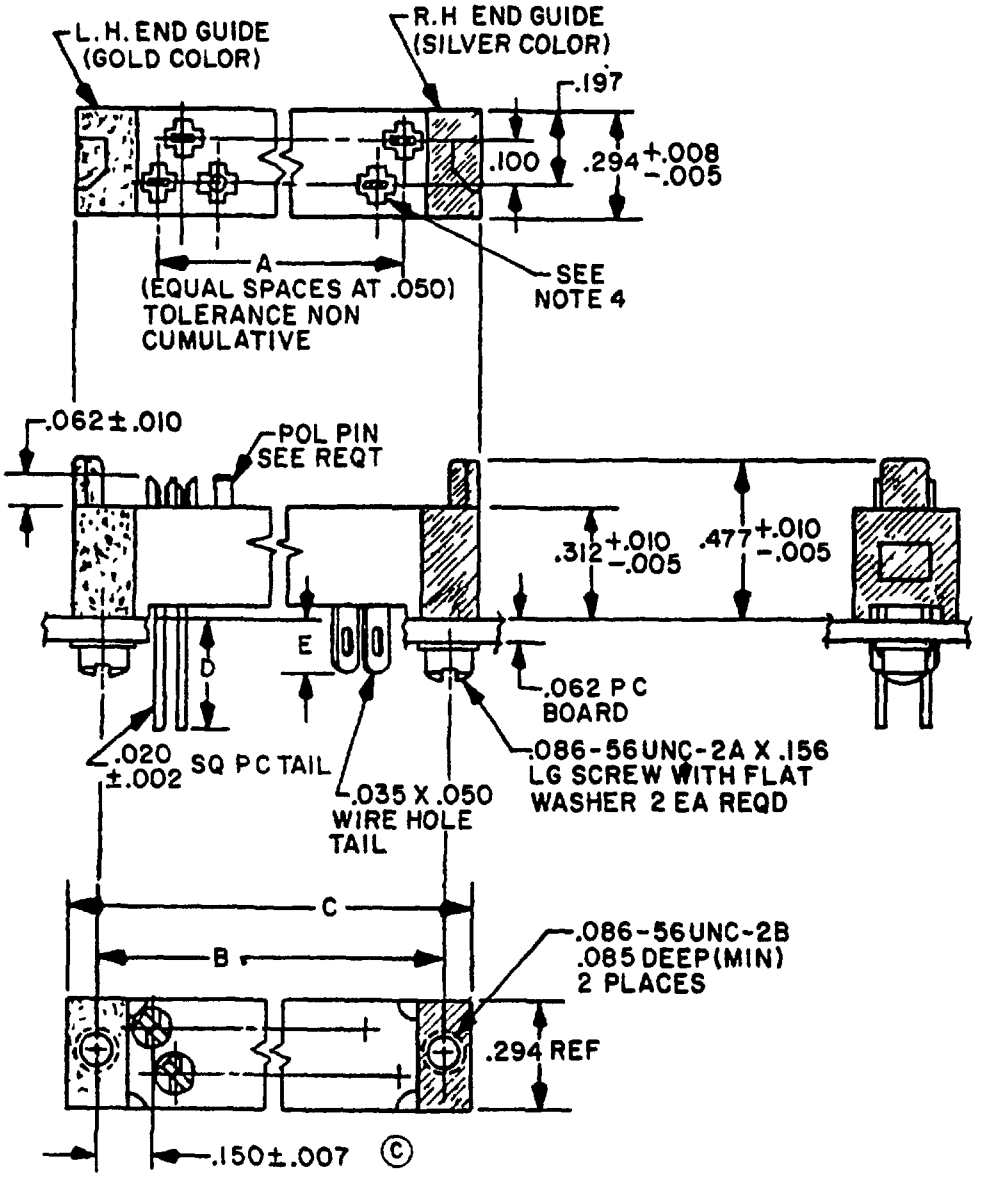
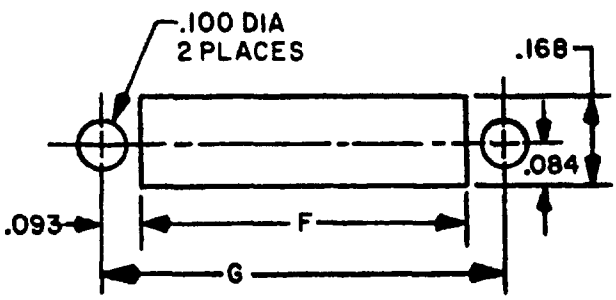


FIGURE 1. Connector, receptacle (.100 spacing).

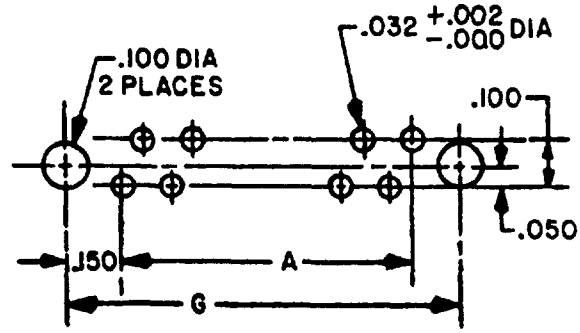
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RECOMMENDED CUTOUT FOR RECEPTACLE USING WIRE HOLE CONTACTS, IN NON-CONDUCTIVE CHASSIS



RECOMMENDED HOLE PATTERN FOR RECEPTACLE USING P C TAIL CONTACTS

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.062	1.57	.187	4.75	.964	24.49	2.200	55.88
.005	.13	.084	2.13	.197	5.00	1.150	29.21	2.350	59.69
.007	.18	.085	2.16	.250	6.35	1.300	33.02	2.500	63.50
.008	.20	.086	2.18	.281	7.14	1.450	36.83	2.650	67.31
.010	.25	.093	2.36	.294	7.47	1.564	39.73	2.764	70.21
.015	.38	.100	2.54	.312	7.92	1.750	44.45	2.950	74.93
.020	.51	.125	3.18	.477	12.12	1.864	47.35	3.100	78.74
.032	.81	.150	3.81	.484	12.29	1.900	48.26	3.550	90.17
.035	.89	.156	3.96	.562	14.27	2.050	52.07	3.664	93.07
.050	1.27	.168	4.27	.850	21.59	2.164	54.97	3.850	97.79
								4.000	101.60

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. With connectors fully engaged the mating contacts shall have a minimum overlap distance of .062 inches. As a minimum the contacts shall exhibit a wiping action through this distance.

FIGURE 1. Connector, receptacle (.100 spacing) - Continued.

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TABLE I. Dash number and dimensions.

Dash 1/ number	No. of contacts	Dimensions (inches)						
		A	B ±.010	C Ref.	D ±.015	E ±.015	F	G
01	18	.850	1.150	1.300	---	.187	.964	1.150
02					.250	---	---	
03					.281	---	---	
04					.562	---	---	
05					.125	---	---	
06					.484	---	---	
07	30	1 .450	1.750	1.900	---	.187	1.564	1.750
08					.250	---	---	
09					.281	---	---	
10					.562	---	---	
11					.125	---	---	
12					.484	---	---	
13	36	1 .750	2.050	2.200	---	.187	1.864	2.050
14					.250	---	---	
15					.281	---	---	
16					.562	---	---	
17					.125	---	---	
18					.484	---	---	
19	42	2.050	2.350	2.500	---	.187	2.164	2.350
20					.250	---	---	
21					.281	---	---	
22					.562	---	---	
23					.125	---	---	
24					.484	---	---	
25	54	2.650	2.950	3.100	---	.187	2.764	2.950
26					.250	---	---	
27					.281	---	---	
28					.562	---	---	
29					.125	---	---	
30					.484	---	---	
31	72	3.550	3.850	4.000	---	.187	3.664	3.850
32					.250	---	---	
33					.281	---	---	
34					.562	---	---	
35					.125	---	---	
36					.484	---	---	

1/ See requirements for complete part number when polarization is required.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material and finish:

Contact: 0.20 inch thick phosphor bronze per QQ-B-750, composition "A". Gold plate per MIL-G-45204, type II, grade D, class 1, over nickel plating per QQ-N-290, class 2, 30 to 150 microinches in the engagement area for a length of .120 inch minimum (see figure 1). The remainder of contact shall be tin lead .0001 microinch minimum per MIL-P-81728 over nickel plating per QQ-N-290, class 2, 30 to 150 microinches.

Insulator: MIL-M-14, type SDG-F.

End guides: Left hand, sintered brass 90/10, clear chromate per MIL-C-5541, class 1A (gold color). Right hand, sintered brass 90/10, nickel plate 300 microinches per MIL-F-14072 (silver color).

Polarizing pin: Aluminum alloy 2011-T3 with clear chromate, per MIL-C-5541, class 1A.

Mounting screw: Brass, cadmium plated, gold chromate finish.

Flat washer: Stainless steel.

Contact identification: Numerical sequence starting with the closest contact adjacent to the left hand end guide (gold color) shall be contact number 1; contact number 2 would be the next offset contact, etc.

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

Individual contact separation force:

Unplated contacts: One ounce minimum when tested in production with gage shown on MIL-C-55302/97.

Plated contacts: One ounce minimum.

Contact resistance: The contact resistance of all pairs measured shall not exceed 20 milliohms.

Low level circuit: The low level circuit resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms, 60 cycles, ac.

High altitude (70,000 feet): 300 volts rms, 60 cycles, ac.

Current rating, maximum: 5 amperes.

Polarizing pin: When required, one of the following codes shall be added to the dash number:

"P" - specifying location by contact number where polarizing pin shall be inserted; for example, M55302/52-01P17 (polarizing pin inserted in place of contact number 17).

"H" - specifying location by contact number where contact shall be omitted for mating; for example, M55302/52-01H17 (polarizing hole has contact number 17 omitted).

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Mating plug: Shall conform to MIL-C-55302/53, /54, /100, /101, /103, and /104.

Part number: M55302/52- (dash number from table I).

Patent number 2,750,572: The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV
Navy - MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3365)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-55302/52C	Connectors, Printed Circuit

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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